

would result in a low device resistance when the magnetization of the two magnets are parallel but when their alignment is antiparallel the resistance would be high*.

A couple of years back it has been envisioned to mimick the same device but entirely with a supramolecular structure. One of the first proposal discussed the possibility to control preferentially spin currents when these where injected through porphyrins adsorbed on a single CNT [257].

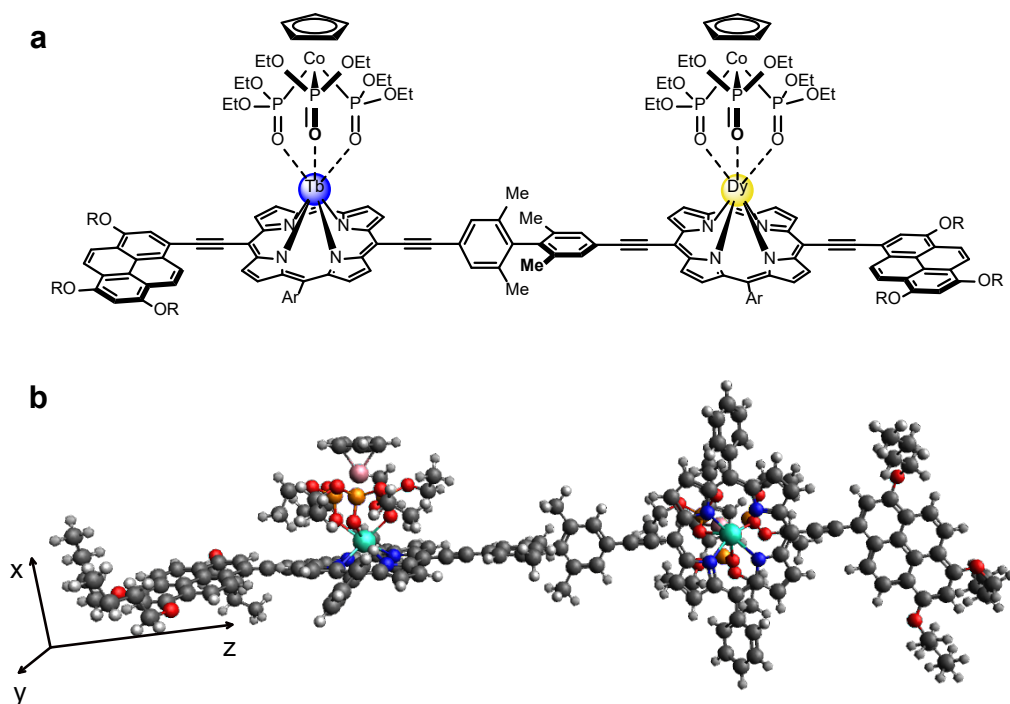


Figure A.1: **a**, Molecular structure of the Tb–Dy porphyrin. In blue and white are highlighted the rare-earth metal centres. $R = -C_{12}H_{25}$; $Ar = 3,5\text{-bis(trihexylsilyl)phenyl}$. **b**, Tight-binding optimised geometry. The $-R$ groups have been discarded to simplify the geometry, and since they do not impact any electronic or magnetic properties. The total number of atoms for this structure is $N = 405$ atoms. Reference system refers to the orientations of the 3D vector magnet in the dilution refrigerator.

The aim of the project is to observe magnetoresistance for a double-porphyrin supramolecular complex, thus demonstrating spin valve capabilities. This requires, first, to establish if the device conductance is suppressed for at certain field value. Since it was not possible to study the individual building blocks of the double

*This argument is strictly valid in case the distance between the two magnets is smaller than the spin-diffusion length $\lambda_s \sim \log\left(\frac{P}{P_0}\right)^{-1}$, and P depends of the spacer length.

porphyrin, we attempted to study the entire molecule.

A.2 Experimental and Computational Methods

Before attempting any transport experiments we carried out an explorative *in-silico* analysis to characterise the magnetic properties of the double porphyrin. To this end, a first important step was to define the type of functional to use. In this respect, LDA and GGA functionals are concave and locals/semi-locals, meaning that the charge density is calculated locally, and so they notoriously tend to describe the electron density for locally respected to the true (unknown) functional. This often leads to an underestimate of the ionization potential and to an overestimate of the electron affinity. This weakness of local functionals is of paramount importance to address especially in strongly-correlated system such as this one, where the presence of f -electrons in a delocalised matrix imposes to heavily correct the systematic error generated by local functionals otherwise resulting in inversions when the spinorbitals are filled.

One strategy to overcome this is to use hybrid functionals, which have been introduced since the early 90s and they include, through various scheme, an exact contribution (Hartree–Fock) to a DFT functional [258, 259]. This procedure, however, is notoriously computationally heavy and one must proceed with caution to even run some preliminary tests. Indeed, because the HF involves the computational of four-body integrals, one has to perform $N(N - 1)$ operations to evaluate these integrals. This translates in a computational time complexity of $\mathcal{O}_{\text{DFT}} * \mathcal{O}_{\text{HF}} = N^5$. When the overlap between the bielectronic integrals is small, algorithmic optimizations typically bring the computational complexity time down to $\mathcal{O}(N^{3-3.5})$. Therefore, if the time employed to run a calculation of about $N = 40$ is about 2 days, if then the system size is increased to $N = 400$ the calculation time would be $(400/40)^3 = 1000$ longer, roughly corresponding to 5 years and half. This assuming the best case scenario in which the calculation scales as a power of three.

Preliminary results have been obtained by adopting the PBEh functional which introduce a 42 percent of exact (HF) exchange and implement a series of corrections

which has previously led to reasonable results even in case of big systems ($N \simeq 150$ atoms) [260, 261].

Fig. A.2 shows the structure of the double porphyrin optimised at DFT level. The molecule was first reduced in dimension by removing the majority of the anchoring group attached at the two terminal position. The total number of atoms was reduced from 405 to 298, which still represents an important system to simulate *ab-initio*. In order to keep the initial test time short, we choose a aug-cc-pVDZ-DK basis set.

A.3 Results

A.3.1 Computational

All measurements were carried out at cryogenic temperatures. The sample was dissolved in toluene and presented excellent solubility even without shaking. The solution was kept in the fridge for almost an year and MALDI performed at intervals of ~ 4 months confirmed the structure stability even when kept in solution in the fridge. Solution of $\sim 8 \mu\text{L}$ were dropcasted on top of graphene tunnel junctions as already outline in the preceeding chapters, and transport properties performed in the Oxford Triton 200 Dilution Refrigerator.

Since the energy differences involved between the FM and the AFM configuration are presumably small, one has to first heavily optimise the geometry so there is a very small error on the bonding forces. Unfortunately a crystal structure of this molecule was not available at the time when this project was carried out. A strong polar implicit solvent (acetonitrile, MeCN) was employed in order to avoid any buckling up of the supramolecular structure.

The problem with this system is twofold: on the one hand you want to go and see presumably small energy differences due to the spins of the different pieces of the molecules first the geometry has to be optimised in a pushy way, so there has to be a very small error on the bonding forces. After that you have to do the FM count first, then flip the magnetisation of orbitals to one side and see the energy difference with the AFM structure. And you want to see this both when the anchor molecules are flat and when they are tilted. So in total one account with four cases.

The challenge with this system is twofold: on the one hand, energy differences due to the spins of the different sections of the molecules will be very small, and on the other hand, the geometry must be optimised so there is minimal inaccuracy in the bonding forces. Then, an FM estimation must be performed before flipping the magnetisation of the f -orbitals and comparing the energy difference with the AFM structure. There are a total of four scenarios to examine, as calculations must be conducted both when the anchor molecules are flat and when they are inclined.

The three rare-earth-oxygen bonds provided by the tripodal (Kläui) ligand are not collinear with the nitrogens belonging to the porphyrinic ring and so they force the porphyrin ring to undergo into a structural distortion.

The structure seems to indicate it preferentially orientates the two aryl rings in the middle perpendicular to each other. One should then take the full-optimised geometry, carefully rotate along the middle triple bond and then explore the four different configurations outline above.

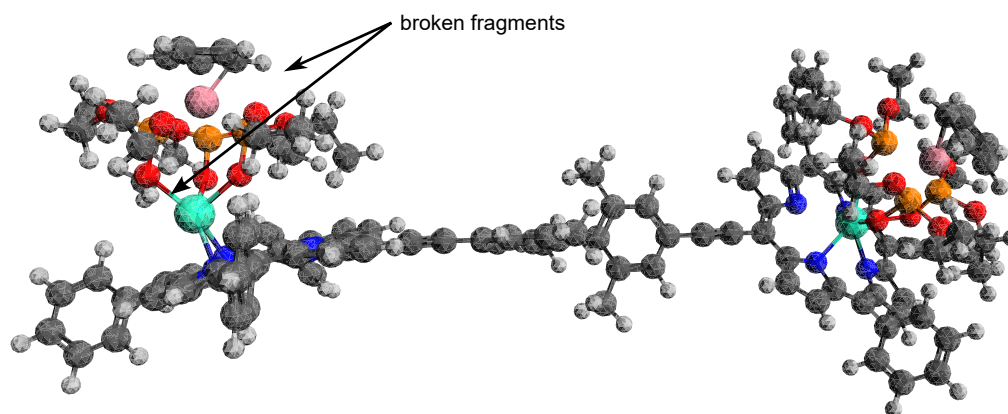


Figure A.2: Tb-Dy structure when optimised at the DFT level. Simulation protocol is detailed in the main text. While the DyP seems to remain intact, the TbP present several critical breaking points in the structure.

After 8 runs of 48h each the structure have started to break down, when the accuracy on the forces was one order of magnitude higher ($\sim 1 \times 10^{-4}$ u) than the one is required to satisfactorily exit the scf cycle. since the system has many degrees of freedom the scaling is not linear to when you get to that limit so

reaching the required level on the force might have taken longer than the time the reached this accuracy level.

The calculation was interrupted as the structure started to break down. Fig. A.2 shows the geometry at the last scf cycle. Breaking of the structure during the geometry optimisation step is often a clear indication that the filling of the spinorbitals is not correct and a detailed study of the two separate porphyrin units is required.

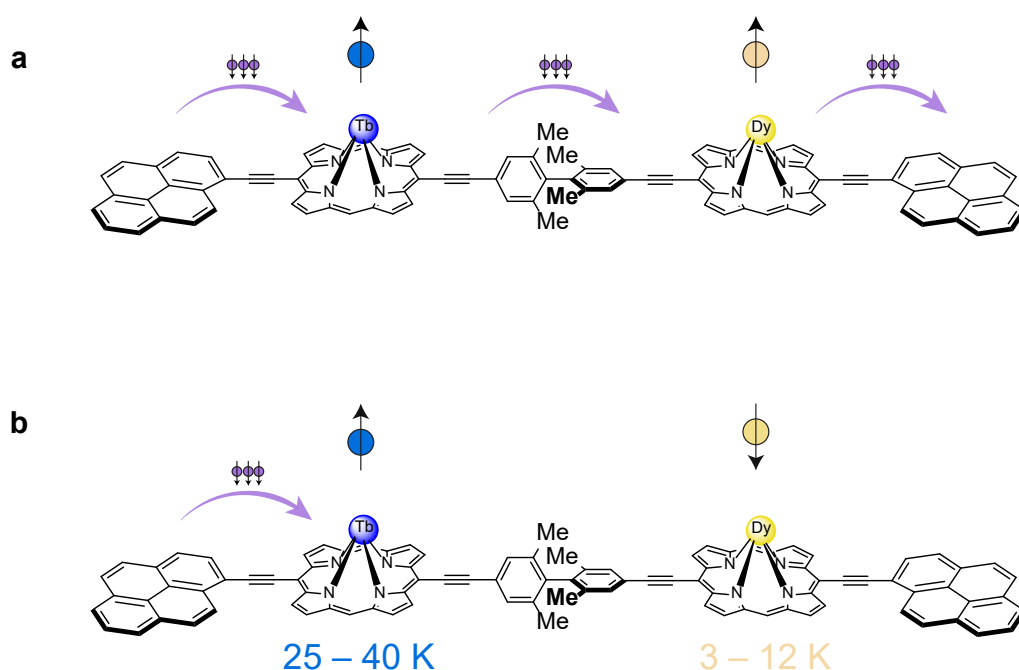


Figure A.3: **a**, When $B = 0$, the Pauli exclusion principles guarantees that a spin-down polarised current may travel across the whole structure resulting in overall low resistance (high G). **b**, When $B \neq 0$, paired spins act as spin blockade and no spin-polarised current can flow across both metals due to the Pauli principle. Indicated are the respective energy splitting amongst the magnetic level for each rare-earth when a magnetic field lifts the degeneracy between spin-up and spin-down states.

A.3.2 Coulomb-Blockade Spectroscopy

We first measured the diamond close zero gate.

We then pushed more forward to study the diamond at -3V. We could unfortunately have not been able to finish the measurements as the device was

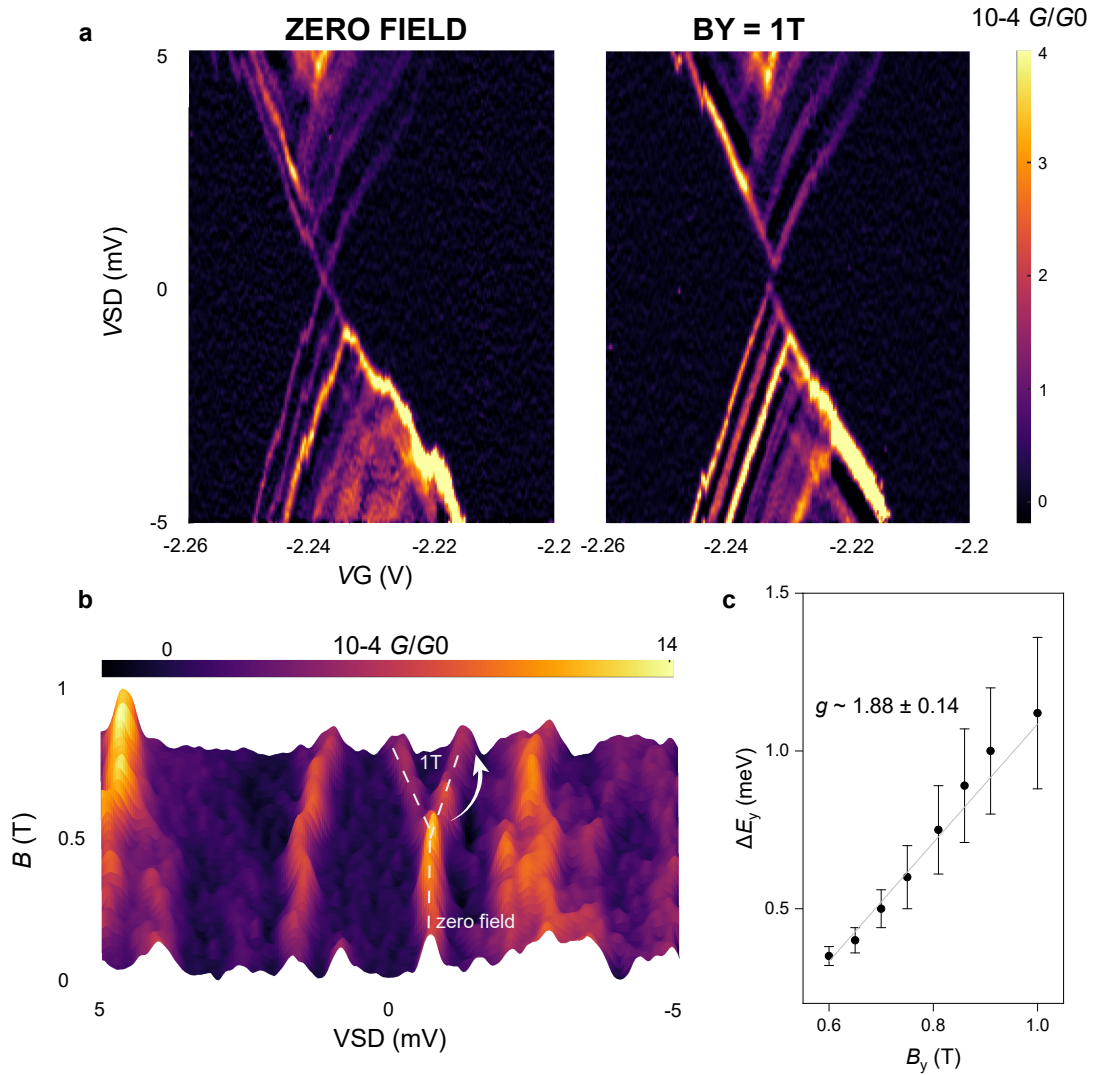


Figure A.4: a, bla b, bla

jumping a lot during the gate and bias scans, possibly due to charge trapped into the HfOx substrate.

A.4 Concluding Remarks

In this chapter we briefly commented the results obtained with DFT and electron transport on Tb–Dy double porphyrin. Due to time limitations we were not able to finish the project, and therefore we cannot present a conclusive outlook. However, we discuss here current problems and future steps.

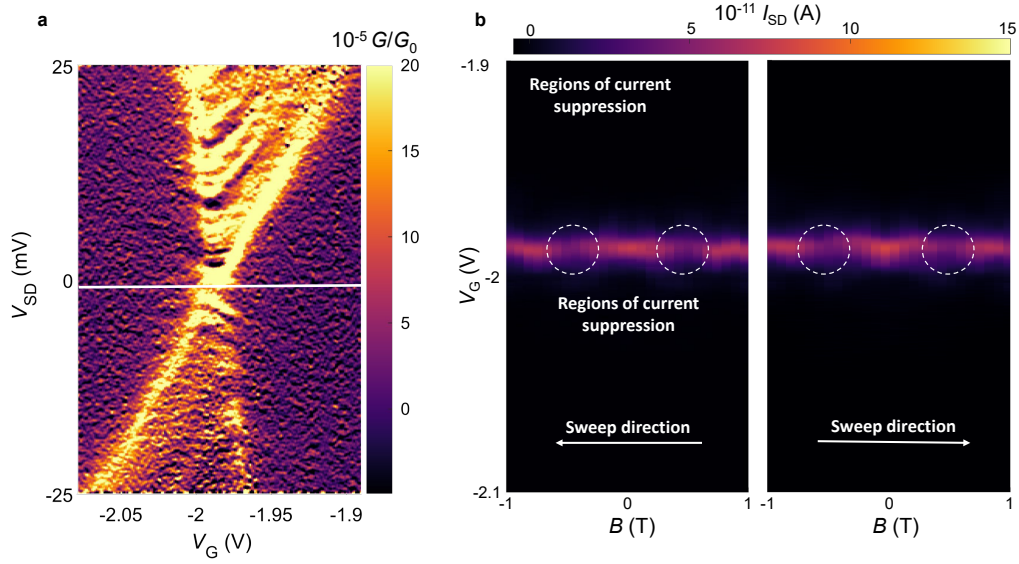


Figure A.5: a, bla b, bla

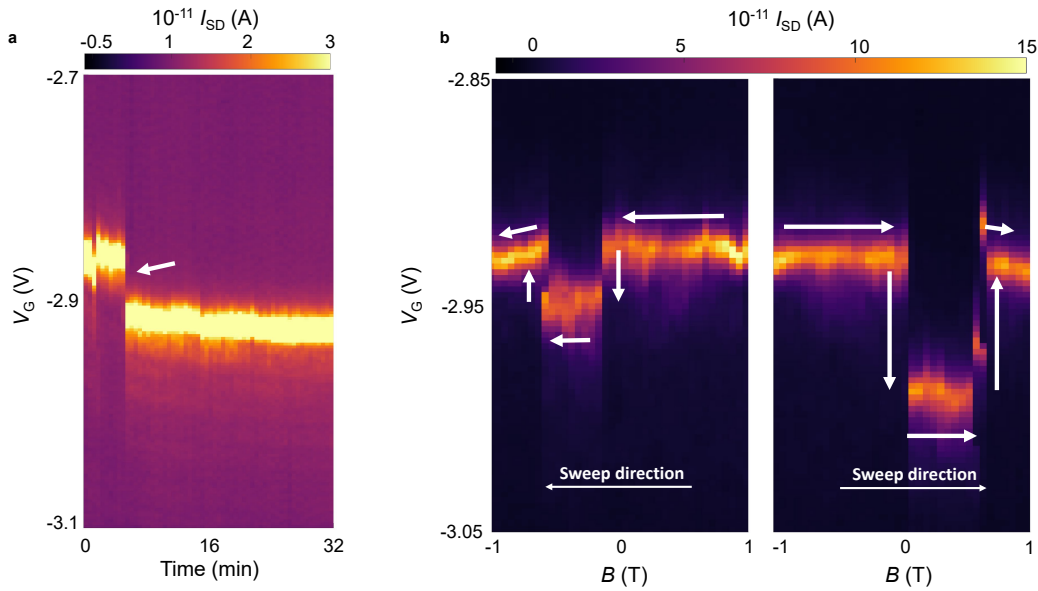


Figure A.6: a, bla b, bla

From a computational perspective, the main factor is the unknown crystal structure, which has forced us to start from a too unoptimised input geometry which have broken during the self-consistent field cycles. From an experimental perspective, the lack of reproducible results might be explained by considering that the preferential orientation of the anchoring group in the ground state seems to be the one with the two aryl units perpendicular to each other, resulting in one of the porphyrin unit to be completely tilted, thus preventing a good anchoring with the graphene leads.

While we are still unable to interpret the spectra and to determine whether or not this molecule display a spin valve behaviour, the raw data show a clear splitting when the magnetic field is swept up to 1 T along the axis coming out from the molecule (y direction). Intuitively, this might be explained by the Zeeman interaction $\mathcal{H}_Z = -\vec{\mu} \cdot \vec{B}$ which couples the Dy magnetic moment with magnetic field. No zero-field splitting of these states are observed which might hint the absence of exchange interactions [262]. Additional information about the energy difference between the FM and the AFM configurations might be obtained by modelling the problem with an Hubbard model as recently performed for simular porphyrin systems [263].

A.5 Project Overview

The Tb–Dy double porphyrin discussed in this project was designed and synthesised by Dr Keith Andrews and prof. Harry Anderson.

B

Sommerfeld Expansion

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Let us consider the following integral:

$$I = \int_{-\infty}^{\infty} h(E) f_0(E) dE, \quad (\text{B.1})$$

where $h(E)$ is a regular function, that it is continuous and differentiable at any order with respect to E , and it is null for $E \rightarrow -\infty$, whereas:

$$f_0(E) = \frac{1}{1 + e^{E-\mu/k_B T}},$$

is the Fermi-Dirac distribution.

B.1 Low-temperature Limit

We demonstrate that, at low temperatures, it is possible to obtain an analytical expression for the integral Eq. B.1 which reads:

$$I = \int_{-\infty}^{\mu} h(E) dE + \frac{\pi^2}{6} (k_B T)^2 [h'(E)]_{\mu} + \frac{7\pi^4}{360} (k_B T)^4 [h'''(E)]_{\mu} + \dots \quad (\text{B.2})$$

For low-temperature here I mean those which satisfy the condition $T \ll T_F$, where $T_F = E_F/k_B$ is the Fermi temperature for an electron gas. Let us define:

$$g(E) = \int_{-\infty}^E h(E) dE.$$

Through an integration per parts, Eq. B.1 becomes:

$$\begin{aligned} I &= \int_{-\infty}^{\infty} h(E) f_0(E) dE = [g(E) f_0(E)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} g(E) \frac{\partial f_0(E)}{\partial E} dE, \\ &= \int_{-\infty}^{\infty} g(E) \left(-\frac{\partial f_0(E)}{\partial E} \right) dE, \end{aligned} \quad (\text{B.3})$$

being the first term null as $g(E) = 0$ for $E \rightarrow -\infty$ and $f_0 = 0$ for $E \rightarrow \infty$. At low temperatures the function $-\frac{\partial f_0}{\partial E}$ has appreciable values only near μ . It is possible to develop a series expansion of $g(E)$ around μ :

$$g(E) = g(\mu) + \sum_{n=1}^{\infty} \left[\frac{(E - \mu)^n}{n!} \right] \left[\frac{d^n g(E)}{dE^n} \right]_{\mu} \quad (\text{B.4})$$

This allow us to write the integral Eq. B.1 as:

$$\begin{aligned} I &= g(\mu) + \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} \frac{(E - \mu)^n}{n!} \left[\frac{d^n g(E)}{dE^n} \right]_{\mu} \left(-\frac{\partial f_0(E)}{\partial E} \right) dE, \\ &= g(\mu) + \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} \frac{(E - \mu)^{2n}}{(2n)!} \left[\frac{d^{2n-1} g(E)}{dE^{2n-1}} \right]_{\mu} \left(-\frac{\partial f_0(E)}{\partial E} \right) dE. \end{aligned} \quad (\text{B.5})$$

where we killed the odd terms in the series as these are integrals of odd functions (note that $\frac{\partial f_0(E)}{\partial E}$ is an even function with respect to E)

By using the change of variable $x = \frac{E - \mu}{k_B T}$, the integral can be written as:

$$I = g(\mu) + \sum_{n=1}^{\infty} c_{2n} (k_B T)^{2n} \left[\frac{d^{2n-1} h(E)}{dE^{2n-1}} \right]_{\mu} \quad (\text{B.6})$$

$$c_{2n} = \frac{1}{(2n)!} \int_{-\infty}^{\infty} x^{2n} \left[-\frac{d}{dx} \frac{1}{e^x + 1} \right] dx, \quad (\text{B.7})$$

Computing the coefficients is done through the integration per parts: by using the following relation:

$$\int_0^{\infty} x^{n-1} \frac{1}{e^x + 1} dx = (1 - 2^{1-n}) \Gamma(n) \zeta(n), \quad (\text{B.8})$$

it is possible to work out a closed form for the coefficients, namely:

$$\begin{aligned} c_{2n} &= \frac{1}{(2n)!} \left[- \left[x^{2n} \frac{1}{e^x + 1} \right]_{-\infty}^{\infty} + 2n \int_{-\infty}^{\infty} x^{2n-1} \frac{1}{e^x + 1} dx \right] \\ &= \frac{1}{(2n)!} \left[0 + 4n (1 - 2^{1-2n}) \Gamma(2n) \zeta(2n) \right] \end{aligned} \quad (\text{B.9})$$

and we obtain

$$\Gamma(n+1) = n! \quad \zeta(2) = \frac{1\pi^2}{6} \quad \zeta(4) = \frac{\pi^4}{90}$$

C

Porphyrin–GNR Hybrid Field-Effect Transistors

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C.1 Additional Data

Here below we report additional data for the pGNRs

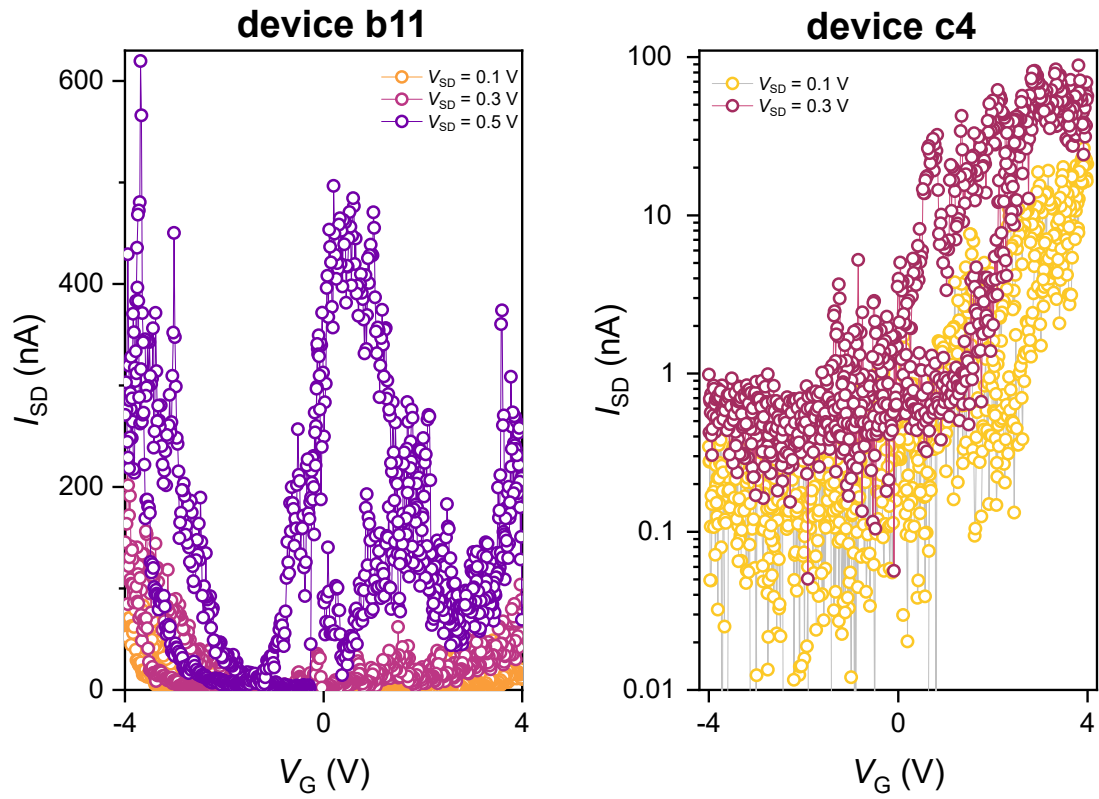


Figure C.1: **left**, FET characteristics: $I_{on}/I_{off} \simeq 10^2$ at $V_{SD} = 0.1$ V, ambipolar behaviour at $V_{SD} = 0.5$ V, $SS \simeq 430$ mV dec $^{-1}$ at $V_{SD} = 0.5$ V. **right**, FET characteristics: $I_{on}/I_{off} \simeq 10^2$ at $V_{SD} = 0.3$ V, small hysteresis present in the $V_{SD} = 0.3$ V transfer characteristics, $SS \simeq 435$ mV dec $^{-1}$ at $V_{SD} = 0.3$ V for the forward sweep, degrades down to $SS \simeq 870$ mV dec $^{-1}$ for the backward sweep, $\mu_{lin} = 11$ cm 2 V $^{-1}$ s $^{-1}$, $\mu_{sat} = 4.3$ cm 2 V $^{-1}$ s $^{-1}$

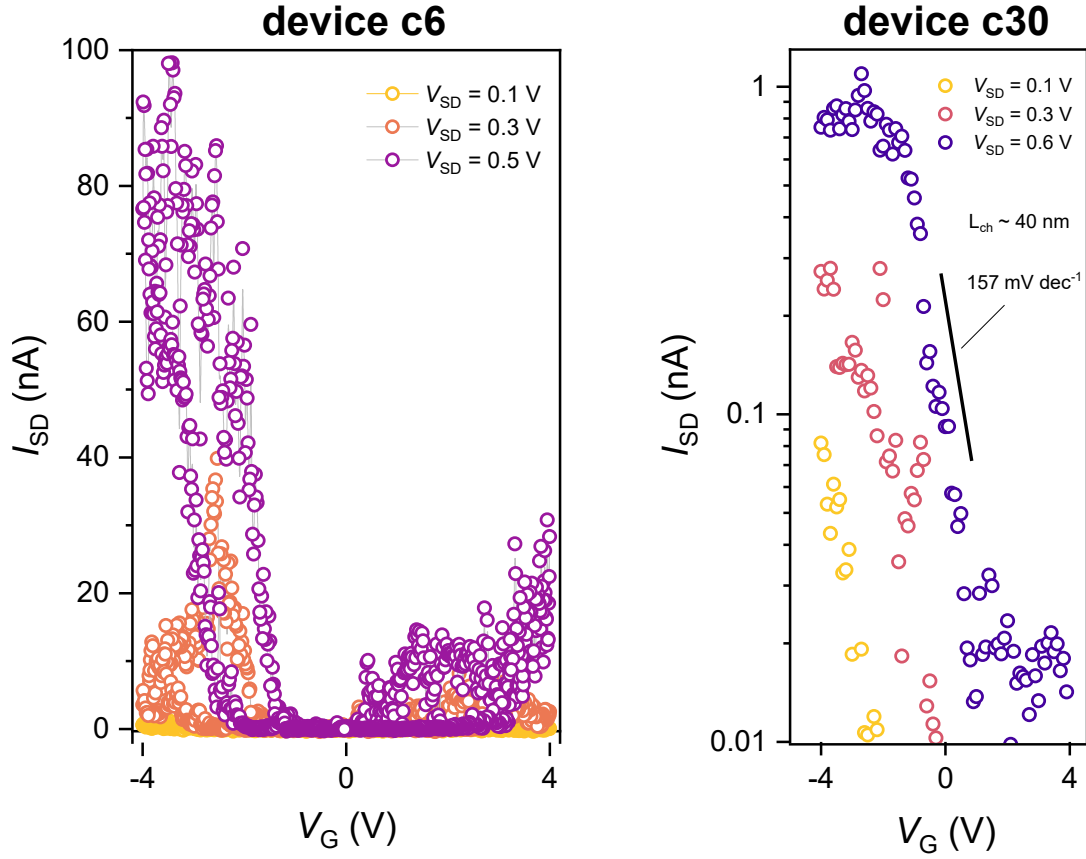


Figure C.2: left, FET characteristics: $I_{on}/I_{off} \simeq 10^2$ at $V_{SD} = 0.3 \text{ V}$, small hysteresis present in the $V_{SD} = 0.3 \text{ V}$ transfer characteristics, gap is a negative gates (slight n-doping), $SS \simeq 860 \text{ mV dec}^{-1}$ at $V_{SD} = 0.5 \text{ V}$ for the forward sweep, degrades down to $SS \simeq 925 \text{ mV dec}^{-1}$ for the backward sweep, $\mu_{lin} = 39 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $\mu_{sat} = 4 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. **right**, FET characteristics: $I_{on}/I_{off} \simeq 10^2$ at $V_{SD} = 0.1 \text{ V}$, $SS \simeq 157 \text{ mV dec}^{-1}$ at $V_{SD} = 0.6 \text{ V}$, estimated channel length $L_{ch} \simeq 40 \text{ nm}$, $\mu_{lin} = 28 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $\mu_{sat} = 6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$

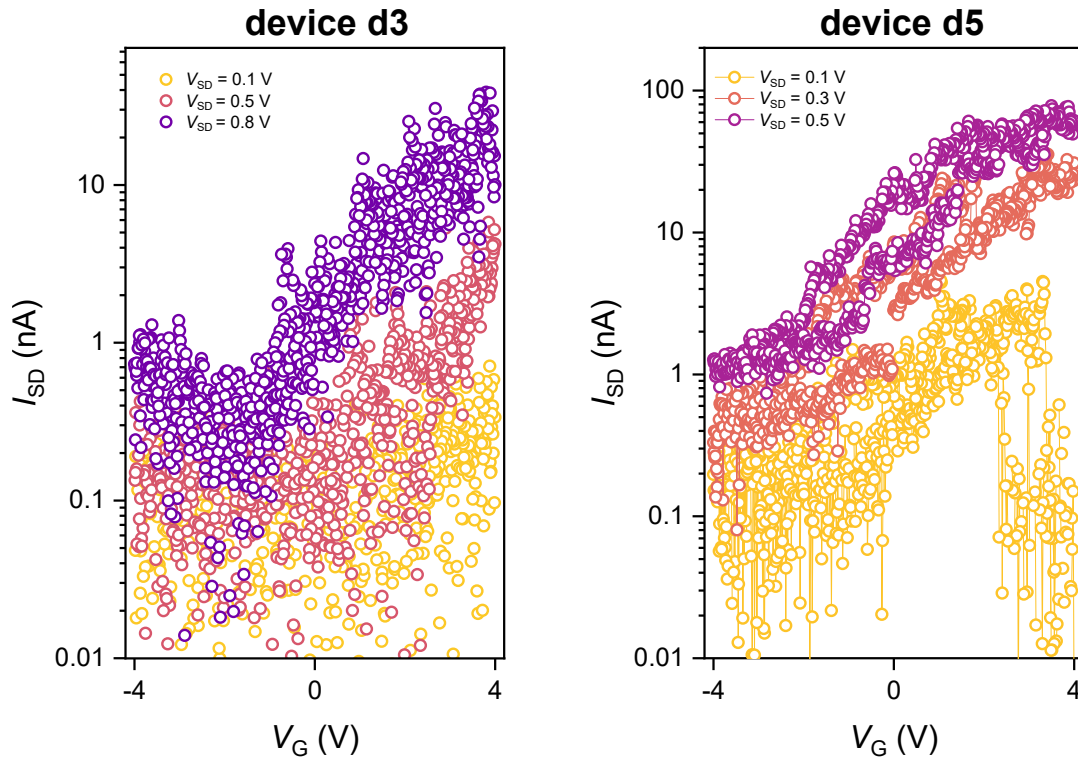


Figure C.3: left, FET characteristics: $I_{on}/I_{off} \simeq 10^1$ at $V_{SD} = 0.1$ V, $SS \simeq 3$ V dec $^{-1}$ at $V_{SD} = 0.6$ V, slightly n-doped, poor off ($I_{off} \simeq 0.1$ nA) and on ($I_{on} \simeq 5$ nA) currents. **right**, FET characteristics: $I_{on}/I_{off} \simeq 10^2$ at $V_{SD} = 0.5$ V, $SS \simeq 1.5$ V dec $^{-1}$ at $V_{SD} = 0.3$ V, n-doped, poor off ($I_{off} < 1$ nA) and on ($I_{on} \simeq 15$ nA) currents.

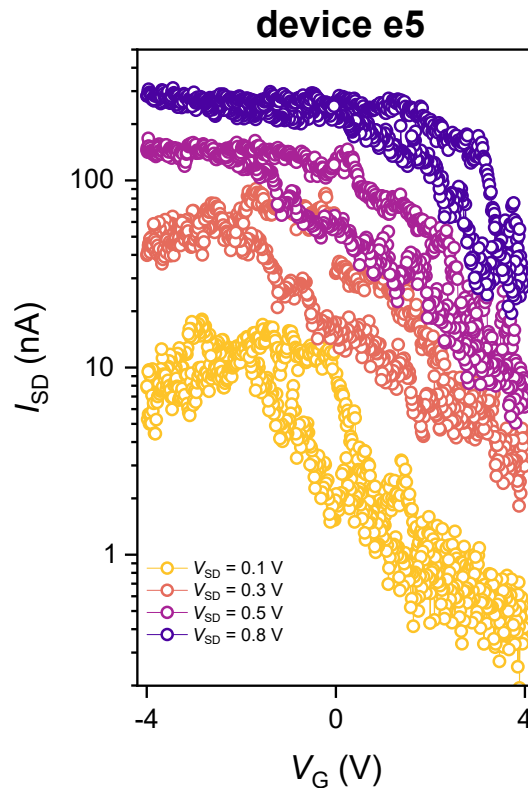


Figure C.4: FET characteristics: $I_{\text{on}}/I_{\text{off}} \simeq 10^2$ at $V_{\text{SD}} = 0.1 \text{ V}$ ($I_{\text{on}}/I_{\text{off}} \simeq 10^1$ at $V_{\text{SD}} = 0.8 \text{ V}$), small hysteresis present in all the transfer characteristics, $SS \simeq 400 \text{ mV dec}^{-1}$ at $V_{\text{SD}} = 0.8 \text{ V}$, $\mu_{\text{lin}} = 5 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $\mu_{\text{sat}} = 1 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$

*The first kind of intellectual and artistic personality
belongs to the hedgehogs, the second to the foxes ...*

— Sir Isaiah Berlin [berlin_hedgehog_2013]

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